

Database Systems Project Proposal

Project Name

IS CAP : Influencer Sponsorship & Campaign Analytics Platform

Roles & Members

Role	Member
Project Lead	Mitchel Vore
Database Designer	Ethan Childs
Query Engineer	Nolan Jones

Problem Statement

Too often there are difficulties with bringing new products to market. Whether the product is sold by a big corporation with millions budgeted for advertising or by an individual with \$500 in savings and genuinely innovative useful product. Our mission is to create a service which will accurately match companies big and small with reputable individuals or groups in order to facilitate the dissemination of products to those individuals which will most readily benefit.

Using systems such as excel, sheets or other manual systems tends to lead to system failures, either creating duplicates that shouldn't exist or bad information being placed due to user error. constraints such as contract per accepted application, tracking multiple deliverables per contract, monitoring engagement metrics over time, and ensuring accurate financial reporting become impossible to enforce and evaluate consistently and very difficult to correct once a mistake has been made.

This project proposes a relational database system that models the full lifecycle of influencer marketing campaigns. The system will enforce meaningful constraints, maintain data integrity, and support advanced analytical queries to evaluate campaign effectiveness, influencer performance, and return on investment.

Requirements and Target Use

Brands: Create and manage sponsorship campaigns, review influencer applications, issue contracts, track deliverables, and monitor campaign performance metrics.

Influencers: Create profiles, apply to campaigns, manage active contracts, submit deliverables, and track earnings and engagement statistics.

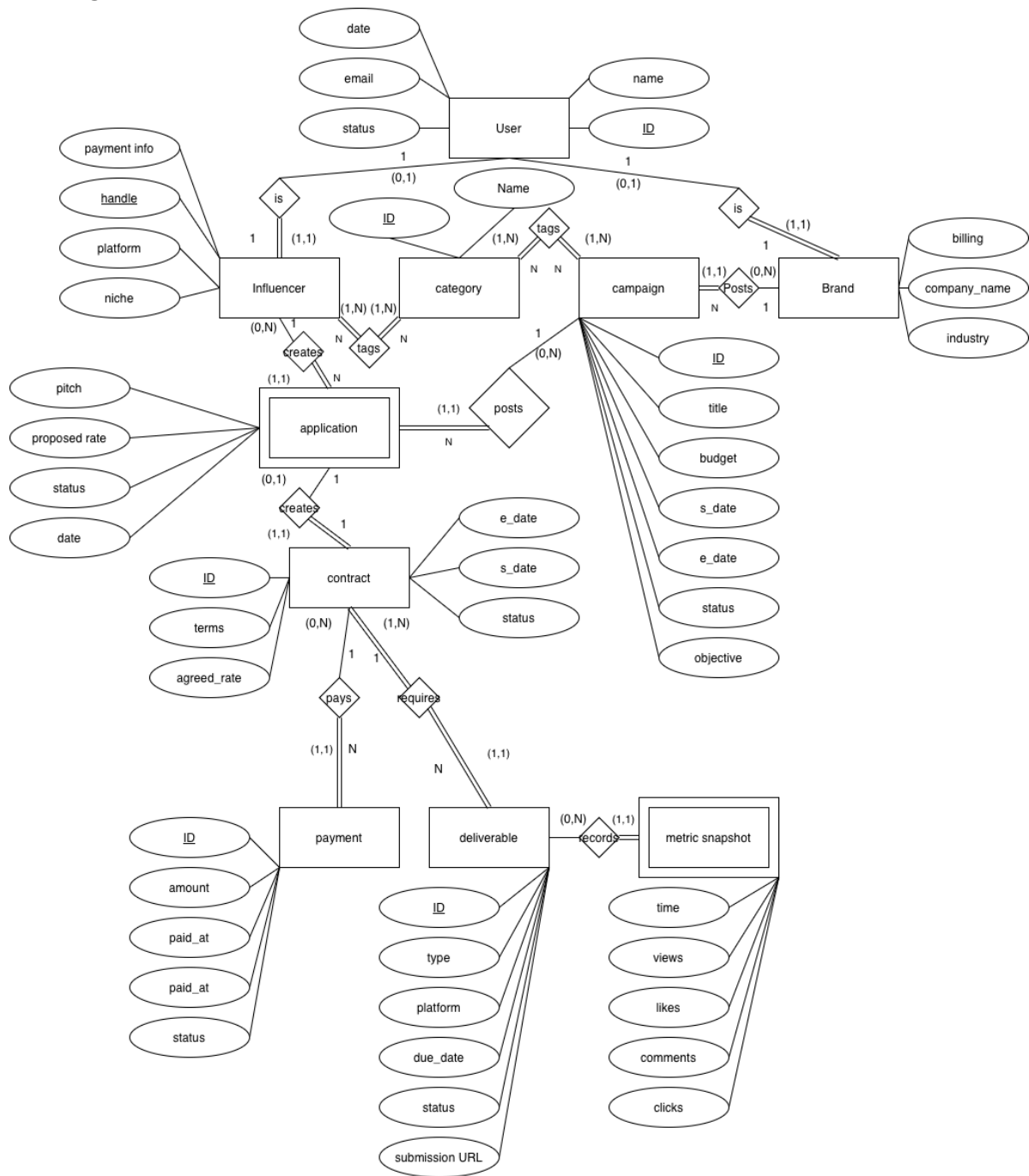
Administrators: Oversee platform activity, resolve disputes, and monitor overall system health.

Core functional requirements include:

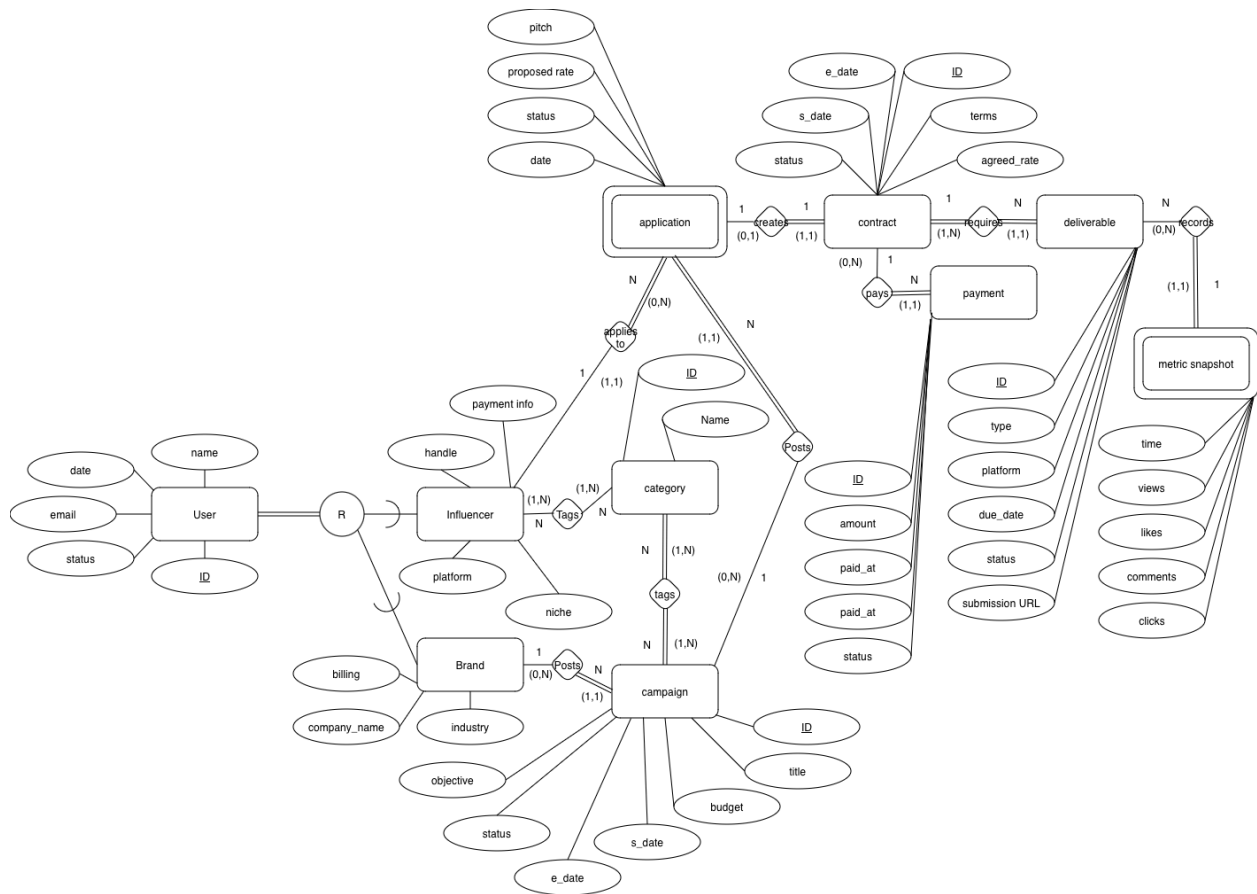
- Ability for brands to create campaigns with defined budgets, timelines, and objectives.
- Influencers can apply to multiple campaigns, and brands can review multiple applications.
- Accepted applications generate formal contracts with defined deliverables.
- Contracts may include multiple deliverables (e.g., posts, videos, stories).
- Engagement metrics (views, clicks, likes, conversions) must be recorded over time.
- Payments must be tracked and linked to contracts.
- The database must support analytical queries such as ROI calculation, top-performing influencers, and campaign performance trends.

Diagrams and Tables

ER Diagram



EER Diagram



Tables

User

<u>User ID</u>	name	start_date	email	account_status	user_type
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Primary Keys : User ID

User_ID : ID for the user : hex : auto increment : non-null

name : First and Last Name : string : no default : non-null

start_date : date users account was created : integer/Date : current date : non-null

email : users email : string : no default : non-null

account_status : if account is activated, banned, locked... : int : 0 : non-null

user_type : will be used to identify as as influencer or brand exclusively : integer : no default : non-null

Influencer

<u>User ID</u>	<u>handle</u>	payment_info	niche	<u>platform</u>
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Primary Keys : handle platform, User ID

Foreign Keys : User ID

handle : stored the users platform handle : string : no default : non-null

payment_info : bank account routing num : long long : no default : non-null

niche : influencers niche : string : null : allows-null

Brand

<u>User ID</u>	company_name	account_to_bill	industry
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Primary Keys : User ID

Foreign Keys : User ID

company_name : the companies name : string : no default : non-null

account_to_bill : company bank acct to bill : long long : no default : non-null

industry : primary industry the company is in : string : no default : non-null

Category

<u>Category ID</u>	name
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Primary Keys : Category_ID

Category ID : ID for the category : hex : auto increment : non-null

name : category name : string : no default : non-null

Campaign

<u>Campaign ID</u>	title	budget	start_date	end_date	status	objective	Brand_ID
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Primary Keys : Category_ID

Foreign Keys : Brand_ID

Campaign ID : ID for the campaign : hex : auto increment : non-null

title : campaign title : string : no default : non-null

budget : budget : float : no default : non-null

start_date : planned start date of campaign : int/Date : no default : non-null

end_date : planned end date of campaign : int/Date : no default : non-null

status : numerical stage representation : int : 0 : non-null

objective : ultimate goal of campaign : string : no default : allows-null

Application

<u>User ID</u>	<u>Campaign ID</u>	pitch	proposed_rate	status	date
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Primary Keys : User_ID Campaign_ID

Foreign Keys : User_ID, Campaign_ID

pitch : what the company wants advertised possibly step by step : string : no default : non-null

proposed_rate : the amount the company wishes to pay individual : float : no default : non-null

status : application status : int : 0 : non-nul

date : date of application submission : int/Date : current date : non-null

Contract

<u>Contract ID</u>	terms	agreed_rate	start_date	end_date	status	iUser_ID	bUser_ID
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Primary Keys : Contract_ID

Foreign Keys : iUser_ID, bUser_ID

terms : terms and conditions : string : no-default : non-null

agreed_rate : payment rate per completed activity : float : no-default : non-null

start_date : date of first commitment : int/Date : no default : non-null

end_date : date of last commitment : int/Date : no default : non-null

status : contract terminations / has ended : 0 : non-null

Payment

<u>Payment ID</u>	amount	paid_at	method	status	Contract_ID
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Primary Keys : Payment_ID

Foreign Keys : Contract_ID

Payment ID : ID for the payment : hex : auto increment : non-null

amount : quantity paid : float : no-default : non-null

paid_at : date of payment : int/Date : no-default : non-null

method : how funds were transferred : string : direct-deposit : non-null

status : unpaid, processing, paid: int : 0 : non-null

Deliverable

<u>Deliverable ID</u>	type	platform	due_date	status	submission_URL	Contract_ID
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Primary Keys : Deliverable ID

Foreign Keys : Contract_ID

Deliverable_ID : ID for the deliverable : hex : auto increment : non-null

type : type of deliverable image, video, url : string : no default : non-null

platform : which platform to complete on : string : no default : non-null

due_date : date to complete by/on : int/Date : no default : non-null

status : completed/missed/tbd : int : 0 ; non-null;

submission_URL : location to upload proof of completion : string : null : allows-null

Metric snapshots

<u>Deliverable ID</u>	<u>time</u>	views	likes	comments	clicks
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Primary Keys : Deliverable_ID time

Foreign Keys : Deliverable_ID

time : date/time of snapshot : int/Date : current time : non-null

views : number of views : int : no default : non-null

likes : number of likes : int : no default : non-null

comments : number of comments : int : no default : non-null

clicks : number of interactions with comments : int : no default : non-null

Influencer_Tags

<u>User ID</u>	<u>Category ID</u>
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Primary Keys : User_ID Category_ID

Foreign Keys :User_ID, Category_ID

Campaign_Tags

<u>Campaign ID</u>	<u>Category ID</u>
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Primary Keys : User_ID Campaign_ID

Foreign Keys : User_ID, Campaign_ID

Data

We will be generating synthetic data and then manually facilitating some data interactions and creation via interfacing.